

Clock generator & Reset fanout

File: clock_gen.kicad_sch

Direct 4-lane PCIe

File: direct_PCIe.kicad_sch

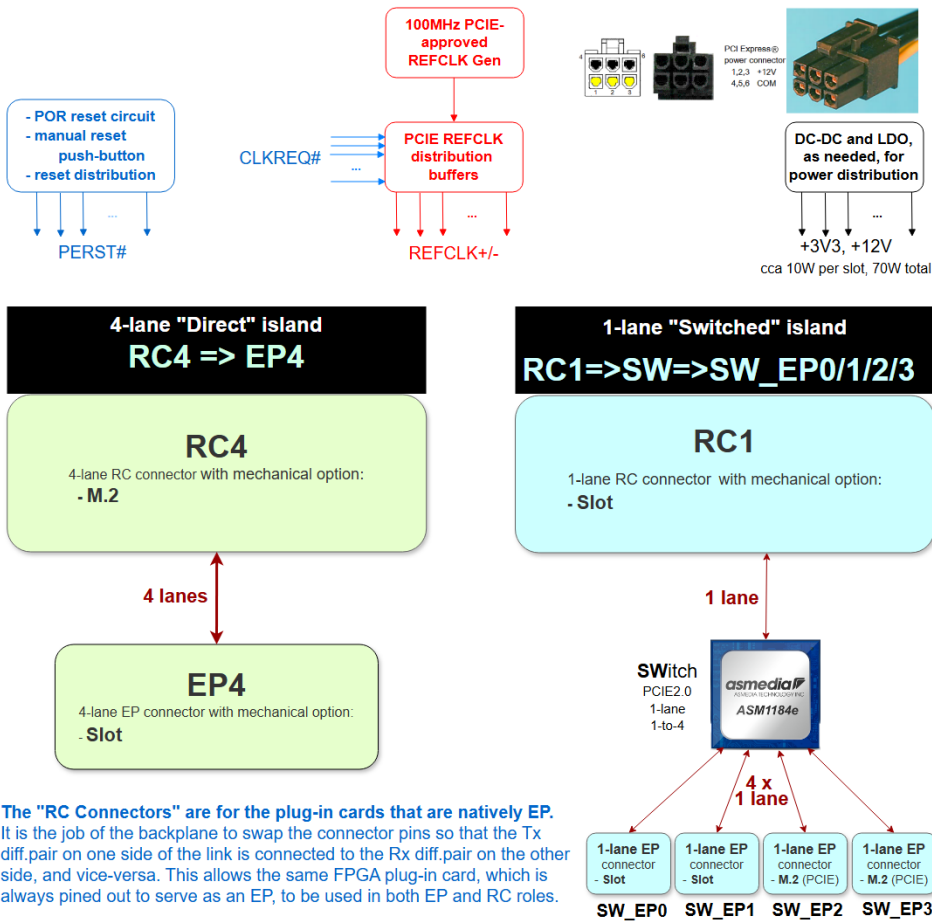
PCIe switch

File: pcie_switch.kicad_sch

Switched 4 x 1-lanes PCIe

File: 4_1-lanes_PCIe.kicad_sch

PCIe mini Backplane

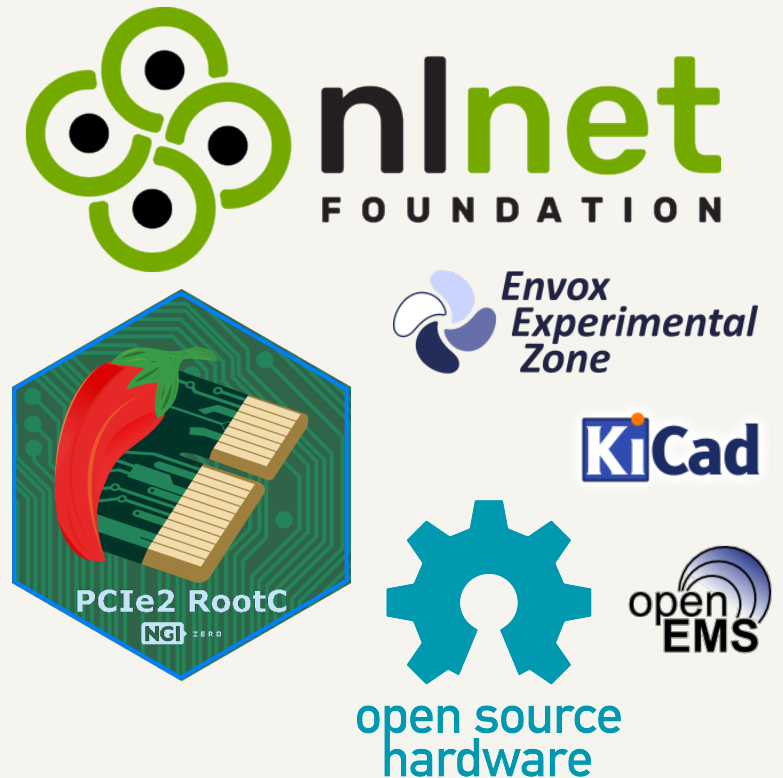


APPROVALS MAY BE VIEWED IN PLM SYSTEM

REV	ECO	DESCRIPTION	DRAWN BY	APPROVED BY	DATE

ELECTRONIC APPROVALS:

ALL APPROVALS FOR THIS DRAWING, ASSOCIATED PARTS LIST(S), AND REVISION CHANGES, ARE CONTROLLED IN THE PRODUCT



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Sheet: /

File: openpci2-backplane.kicad_sch

Title: openPCIe2 Backplane

Size: A4 Date: 2026-10-06

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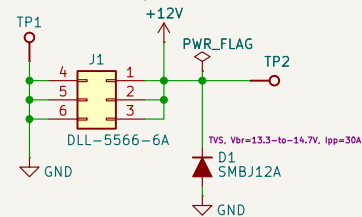
Rev: RevB

Id: 1/5

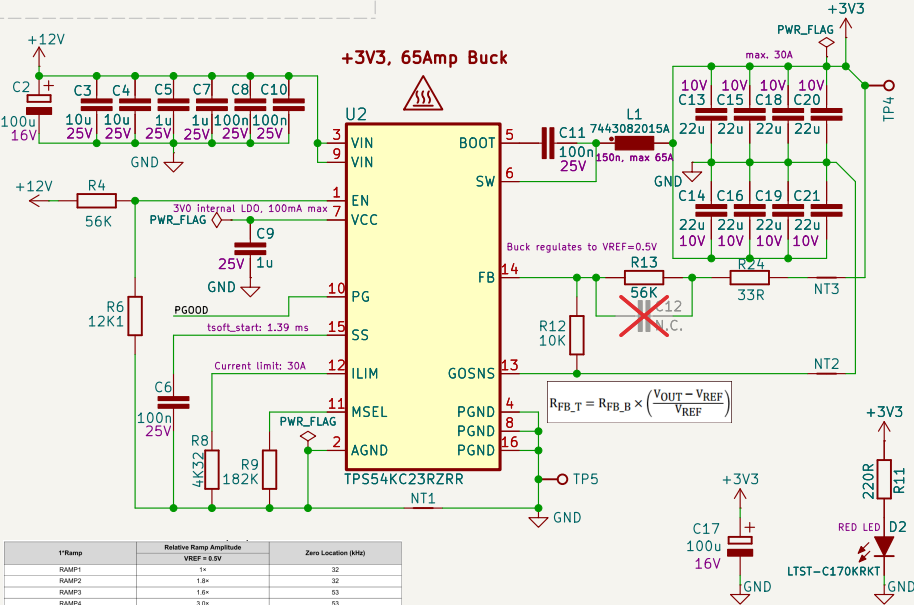


Power Supply

+12V Power input



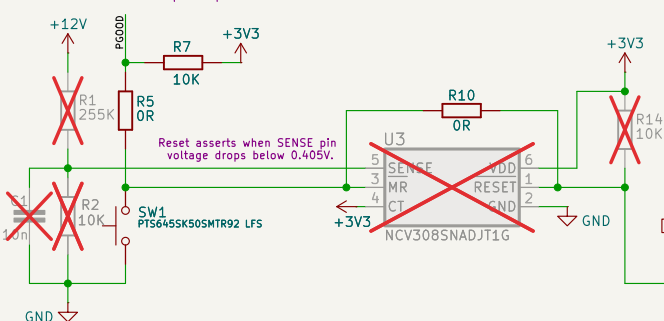
+3V3, 65Amp Buck



V/Ramp	Relative Ramp Amplitude VREF = 0.5V	Zero Location (Hz)
RAMP1	1.0	32
RAMP2	1.0	32
RAMP3	1.0	53
RAMP4	3.0	53

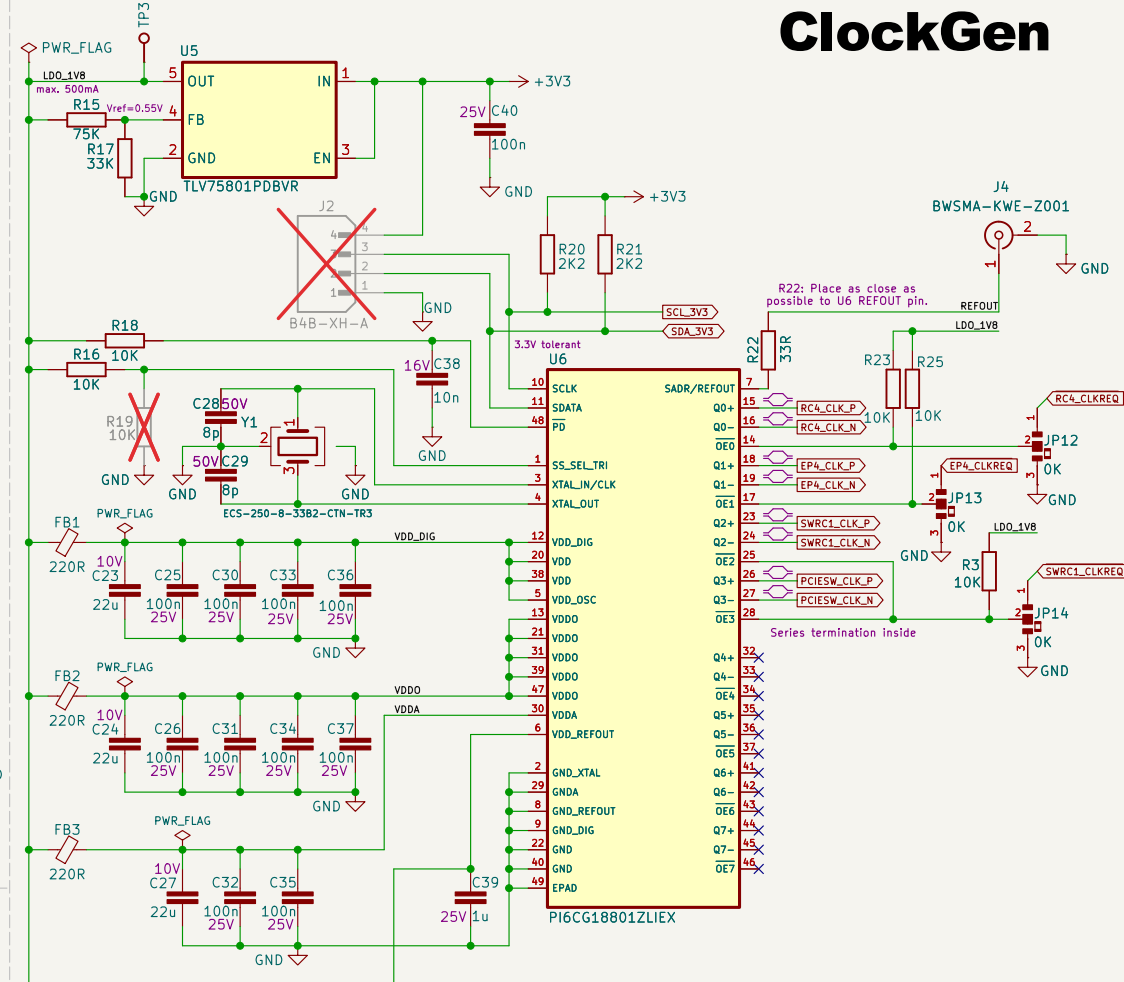
Pin	FB	I	Output voltage feedback input. A resistor divider from the output voltage to GOSNS (tapped to FB pin) sets the output voltage. Connect the FB divider to the output voltage near the load.
14	FB	I	Output voltage feedback input. A resistor divider from the output voltage to GOSNS (tapped to FB pin) sets the output voltage. Connect the FB divider to the output voltage near the load.
13	GOSNS	I	Negative input of the differential remote sense circuit. Connect to a ground sense point near the load.

Reset asserts when +12V input drops below -10.73V.



Voltage Supervisor Manual Reset, Reset Fanout

ClockGen



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Sheet: /Clock generator & Reset fanout/

File: clock_gen.kicad_sch

Title: openPCIE2 Backplane

Size: A4

Date: 2026-10-06

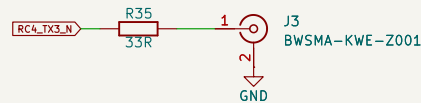
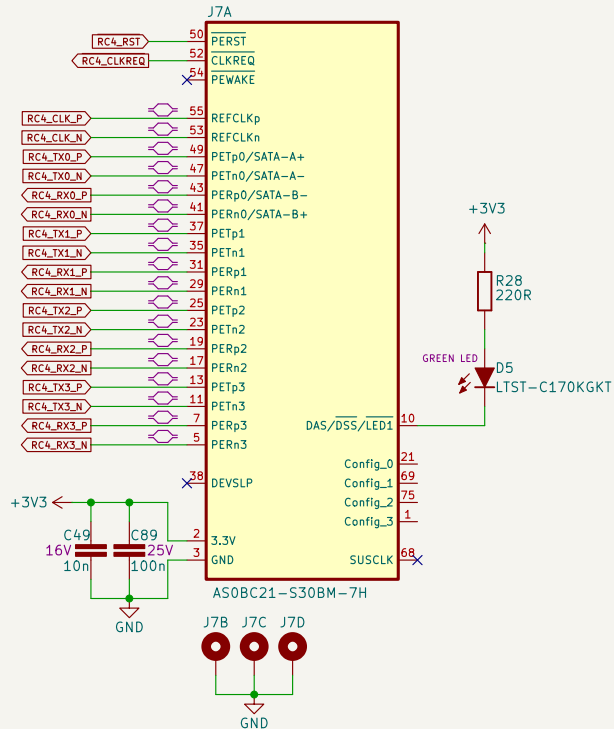
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Rev: RevB

Id: 2/5

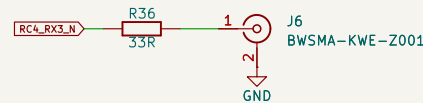
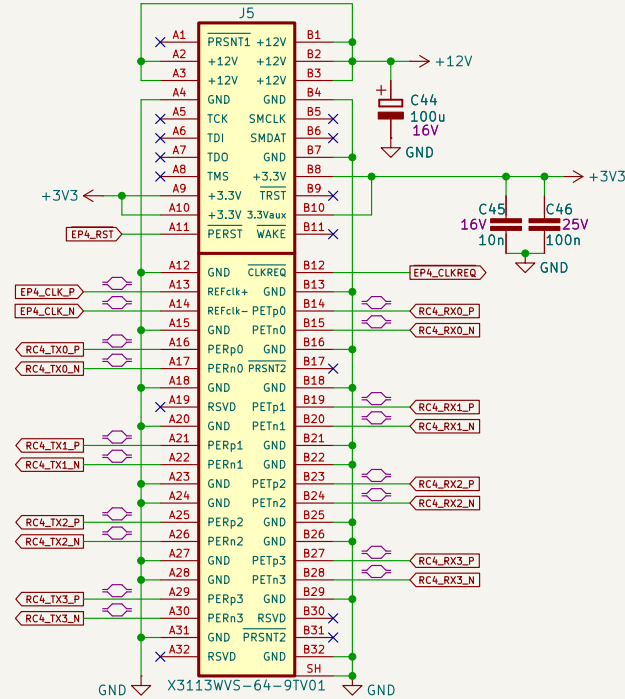
Direct Connection Island

RC4



R and Connector must be placed right next to the trace to eliminate the stub.

EP4



R and Connector must be placed right next to the trace to eliminate the stub.

PCIEx1 pinout

Pin	Side B	Side A	Description
1	+12 V		Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	Main power pins
3	+12 V	+12 V	
4	Ground	Ground	
5			SMBus and JTAG port pins
6			
7	Ground		
8	+3.3 V		Aux power & Standby power
9		+3.3 V	
10	+3.3 V aux	+3.3 V	
11			Link reactivation, fundamental reset [25]
Key notch			
12	CLKREQ#(20)	Ground	Clock Request Signal
13	Ground	REFCLK+	Reference clock differential pair
14	HSOp(0)	REFCLK-	Lane 0 transmit data, + and -
15	HSOn(0)	Ground	
16	Ground	HSIp(0)	Lane 0 receive data, + and -
17		HSIn(0)	
18	Ground	Ground	

PCI Express x1 cards end at pin 18

Legend

Ground pin	Zero volt reference
Power pin	Supplies power to the PCIe card
Card-to-host pin	Signal from the card to the motherboard
Host-to-card pin	Signal from the motherboard to the card
Open drain	May be pulled low or sensed by multiple cards
Sense pin	Tied together on card
Reserved	Not presently used, do not connect

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Sheet: /Direct 4-lane PCIe/

File: direct_PClE.kicad_sch

Title: openPCIe2 Backplane

Size: A4 Date: 2026-10-06

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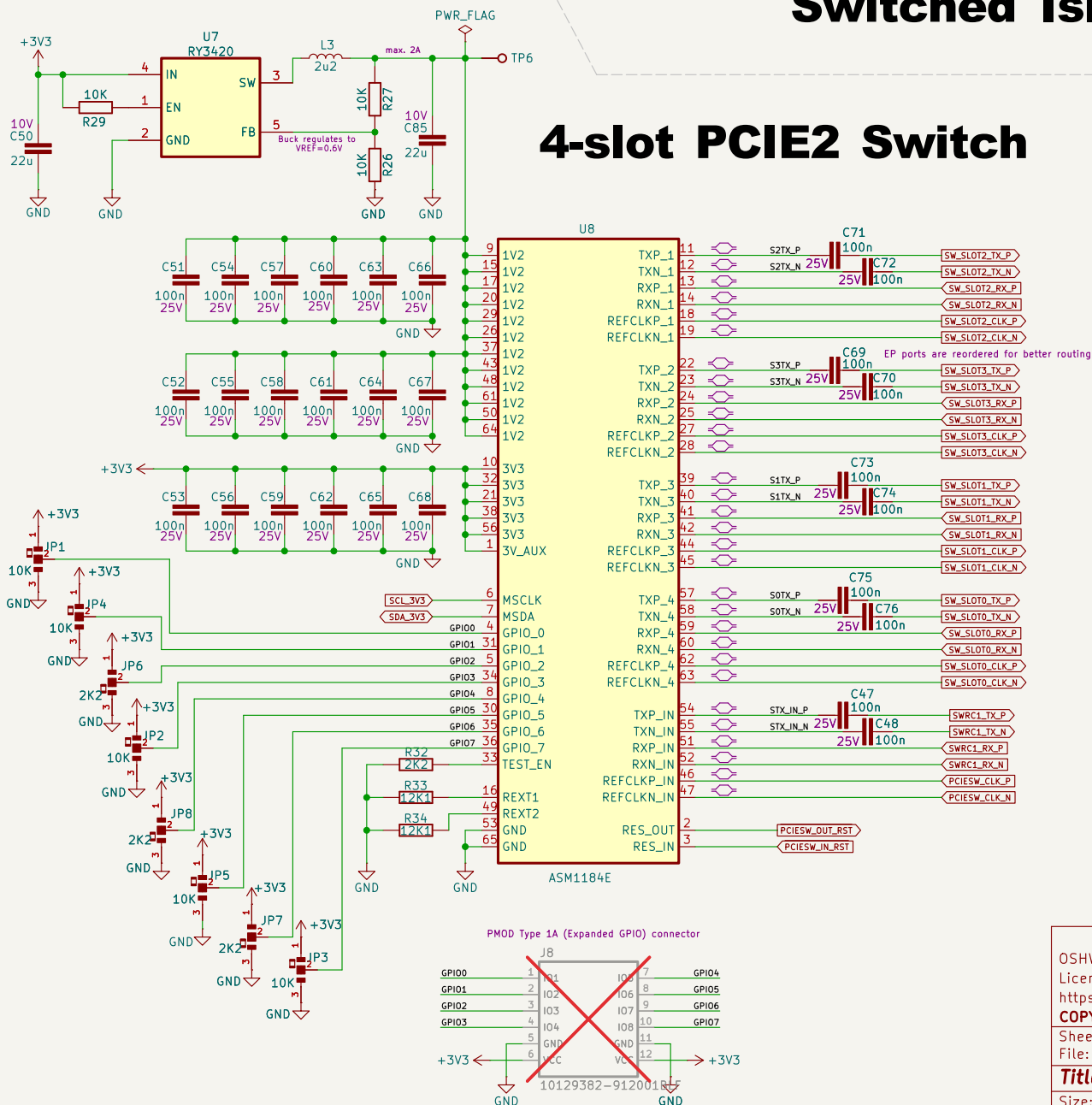
Rev: RevB

Id: 3/5

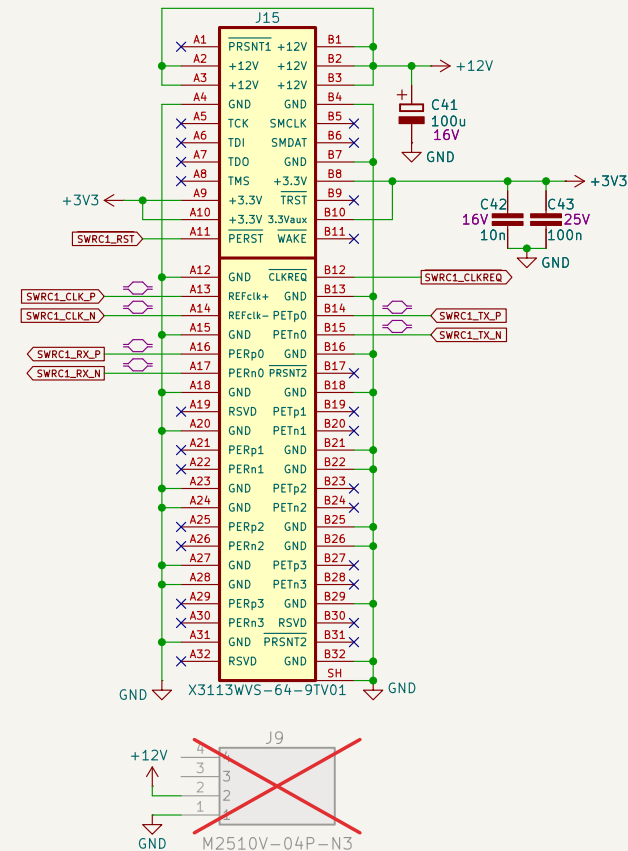


Switched Island (Part 1)

4-slot PCIE2 Switch



SWRC1



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Sheet: /PCle switch/

File: pcie_switch.kicad_sch

Title: openPCIE2 Backplane

Size: A4 Date: 2026-10-06

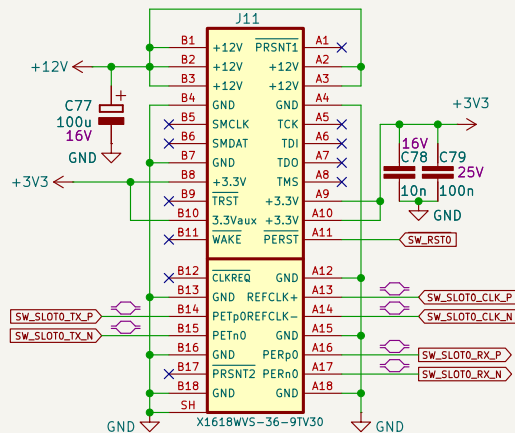
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Rev: RevB

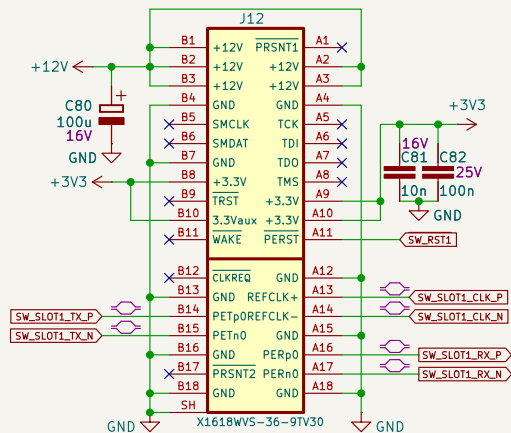
Id: 4/5

Switched Island, Part 2

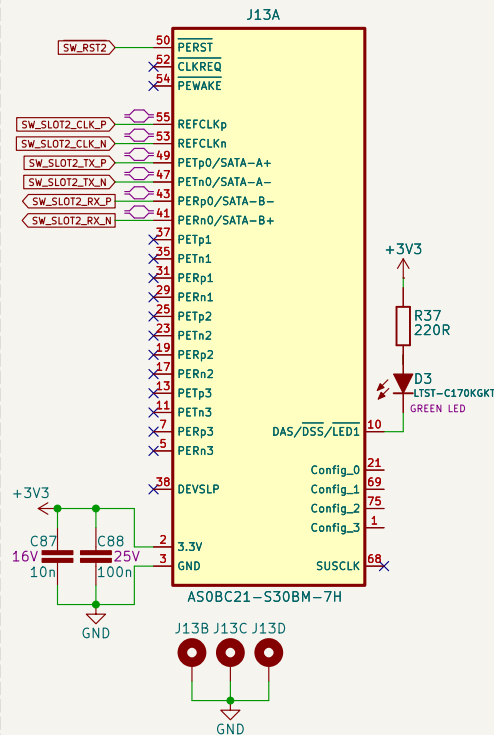
SW_EP0



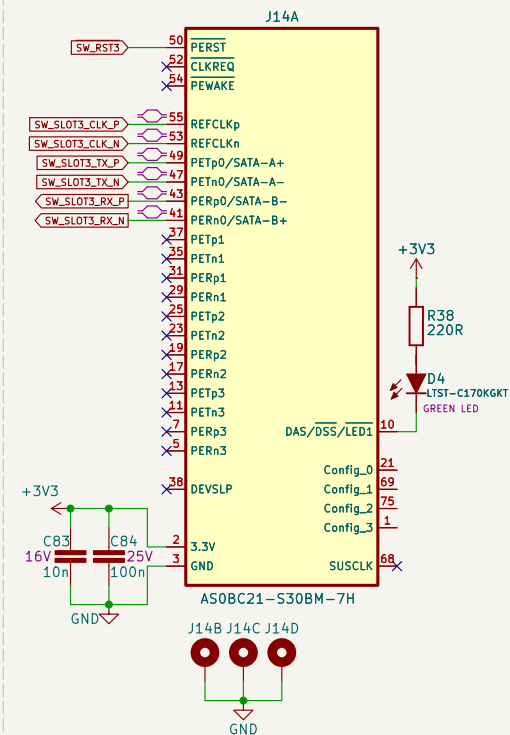
SW_EP1



SW_EP2



SW_EP3



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Sheet: /Switched 4 x 1-lanes PCIe/

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Size: A4 Date: 2026-10-06

KiCad E.D.A. 10.0.6

Rev: RevB

Id: 5/5